

## **ASSIGNMENT**

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**Course Name:** Computer Networks

**Course Code:** CSA0704

**Github URL:** <https://github.com/suri307/Computer-Network/tree/main/ASSIGNMENT>

### **TITLE**

**Design, Configure, Analyse and Evaluate a Multi-Branch Enterprise Network Using IP Addressing, Routing, Protocols and Network Topologies for Reliable, Scalable, and Efficient Inter-Branch Communication**

## **ABSTRACT**

This project presents the design, implementation, and evaluation of a multi-branch enterprise computer network using Cisco Packet Tracer. The proposed network consists of a Head Office and two Branch Offices with multiple departments and end devices. The network is designed to support reliable communication between departments and branches using IPv4 addressing, subnetting, default gateways, routing, switches, routers, and appropriate transmission media.

The implementation includes the configuration of routers, switches, and end devices followed by connectivity and routing verification. IP addressing is assigned to different network segments, and routing tables are examined to verify the availability of remote networks. Ping testing is performed to evaluate connectivity, packet loss, and round-trip time. Cisco Packet Tracer Simulation Mode is also used to observe packet movement and network events.

The project additionally considers star, mesh, bus, and hybrid topologies and evaluates them based on scalability, reliability, cost, fault tolerance, and ease of management. TCP, UDP, HTTP, and DNS are considered for protocol-level analysis under different network conditions. The final design demonstrates how appropriate addressing, routing, topology selection, and performance evaluation can contribute to a scalable enterprise network.

# **CHAPTER 1 – INTRODUCTION**

## **1.1 Background**

Modern organizations depend on computer networks for communication, resource sharing, application access, and information exchange. When an organization operates from multiple locations, the network must support communication between different departments and branches while maintaining reliability and scalability.

A multi-branch enterprise network contains several interconnected LANs. Each LAN may represent a department or branch office and requires appropriate IP addressing and routing. A poorly designed network can result in address conflicts, communication failures, congestion, difficult troubleshooting, and poor scalability.

This project therefore focuses on designing a simulated enterprise network using Cisco Packet Tracer. The assignment requires at least 36 end devices, IPv4 addressing with subnetting/VLSM, default gateways, routing between multiple network segments, and demonstration and comparison of different network topologies.

## **1.2 Problem Statement**

A growing organization has a Head Office and two Branch Offices. The organization contains departments such as:

- Human Resources
- Research and Development
- Sales
- Finance

The organization requires a network that provides:

- Department-level connectivity.
- Branch-to-branch communication.
- Efficient IPv4 address allocation.
- Inter-network routing.
- Reliable packet delivery.
- Easy troubleshooting.
- Scalability for future users.
- Performance monitoring.

The network must be designed, simulated, tested, and evaluated before deployment.

## **1.3 Problem Formulation**

The problem can be formulated as:

**How can a multi-branch enterprise network be designed and configured using appropriate IPv4 addressing, subnetting, routing, topology, devices, cabling, and protocols so that reliable and scalable communication is achieved between departments and branch offices?**

The solution must consider both technical and engineering factors such as:

- Number of users.
- Number of departments.
- IP address requirements.
- Routing requirements.
- Network topology.
- Device selection.
- Protocol behaviour.
- Performance.
- Reliability.
- Cost.
- Scalability.

This directly addresses the rubric's requirement to clearly identify users, departments, hosts, services, addressing requirements, topology requirements, protocol scenarios, and performance objectives.

## **CHAPTER 2 – OBJECTIVES AND EXPECTED OUTCOMES**

### **2.1 Objectives**

The objectives are:

1. To design a multi-branch enterprise network.
2. To connect a Head Office with two Branch Offices.
3. To provide connectivity for at least 36 end devices.
4. To develop an IPv4 addressing scheme.
5. To apply subnetting/VLSM where appropriate.
6. To configure default gateways.
7. To configure routing between network segments.
8. To verify routing tables.
9. To test end-to-end connectivity.
10. To analyse packet loss and latency.

11. To demonstrate star, mesh, bus, and hybrid topologies.
12. To analyse TCP, UDP, HTTP, and DNS behaviour.
13. To evaluate network performance under different conditions.
14. To identify limitations and bottlenecks.
15. To justify major network design decisions.

The rubric expects the project to go beyond simply creating a topology and requires protocol analysis, topology comparison, and performance evaluation.

## 2.2 Expected Outcomes

After implementation, the network should provide:

- Successful communication between configured network segments.
- Correct IP addressing.
- Correct default gateway configuration.
- Working routing between networks.
- Reduced addressing conflicts.
- Measurable packet loss and latency.
- Observable packet movement.
- A scalable network structure.
- A basis for analysing TCP, UDP, HTTP, and DNS.

## CHAPTER 3 – REQUIREMENTS ANALYSIS

### 3.1 User Requirements

The network is designed for users belonging to different departments.

| Location    | Department/Group | Requirement    |
|-------------|------------------|----------------|
| Head Office | HR               | Department LAN |
| Head Office | R&D              | Department LAN |
| Head Office | Sales            | Department LAN |
| Head Office | Finance          | Department LAN |
| Branch 1    | Branch users     | Branch LAN     |
| Branch 2    | Branch users     | Branch LAN     |

The final assignment requires a minimum of 36 end devices.

### 3.2 Network Requirements

The network must support:

- IPv4 addressing.
- Subnetting.
- Default gateways.
- Routing.
- LAN communication.
- Inter-branch communication.
- Protocol communication.
- Network monitoring.
- Fault analysis.
- Performance measurement.

### 3.3 Hardware Requirements

| Device            | Purpose                                            |
|-------------------|----------------------------------------------------|
| Router            | Inter-network communication and routing            |
| Switch            | LAN connectivity                                   |
| PC                | End-user device                                    |
| Copper cable      | Ethernet connectivity                              |
| Serial connection | Router-to-router WAN connectivity where applicable |

### 3.4 Software Requirements

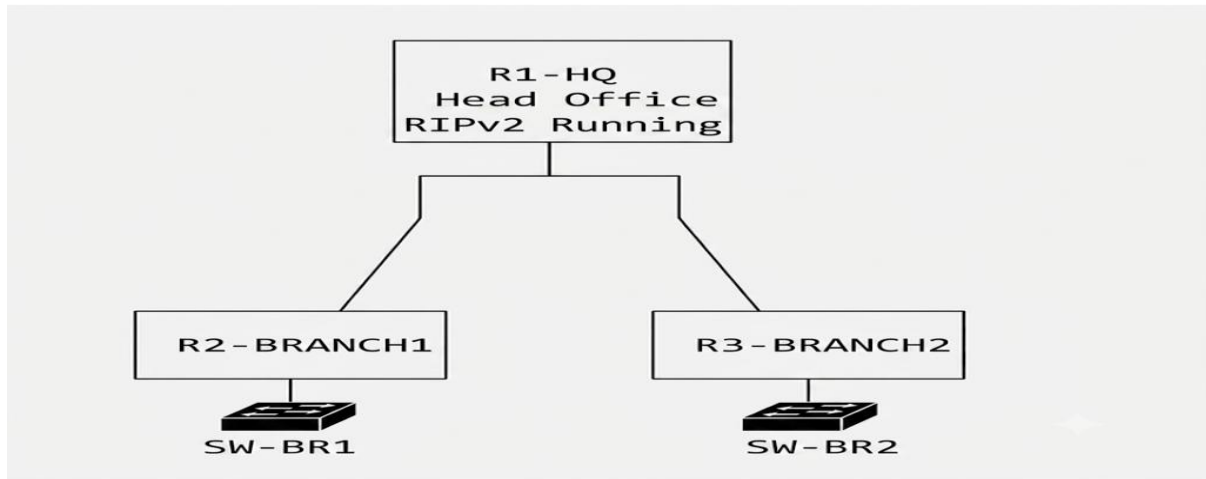
**Cisco Packet Tracer** is used as the primary simulation environment.

The assignment specifically permits Packet Tracer, GNS3, EVE-NG, or equivalent simulation tools.

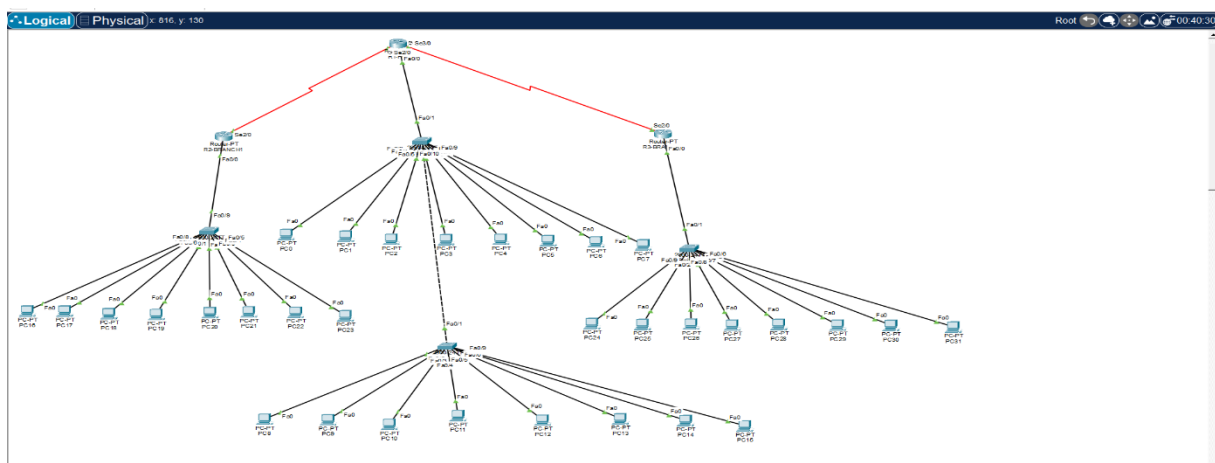
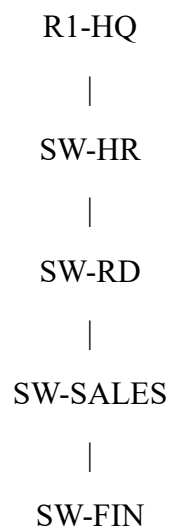
## CHAPTER 4 – NETWORK DESIGN

### 4.1 Proposed Architecture

The network contains three major routing locations:



The Head Office contains departmental switches:



**Figure 4.1:** Proposed Multi-Branch Enterprise Network Topology

## CHAPTER 5 – IP ADDRESSING AND SUBNETTING

This section is particularly important because **IP Addressing, Subnetting & Routing** carries **20 marks**, the highest individual rubric allocation. Level 4 requires correct addressing, subnetting/VLSM, gateways, routing tables, and basic routing configuration with efficient/error-free addressing.

### 5.1 Addressing Strategy

Private IPv4 addresses are used to separate different network segments.

Example allocation:

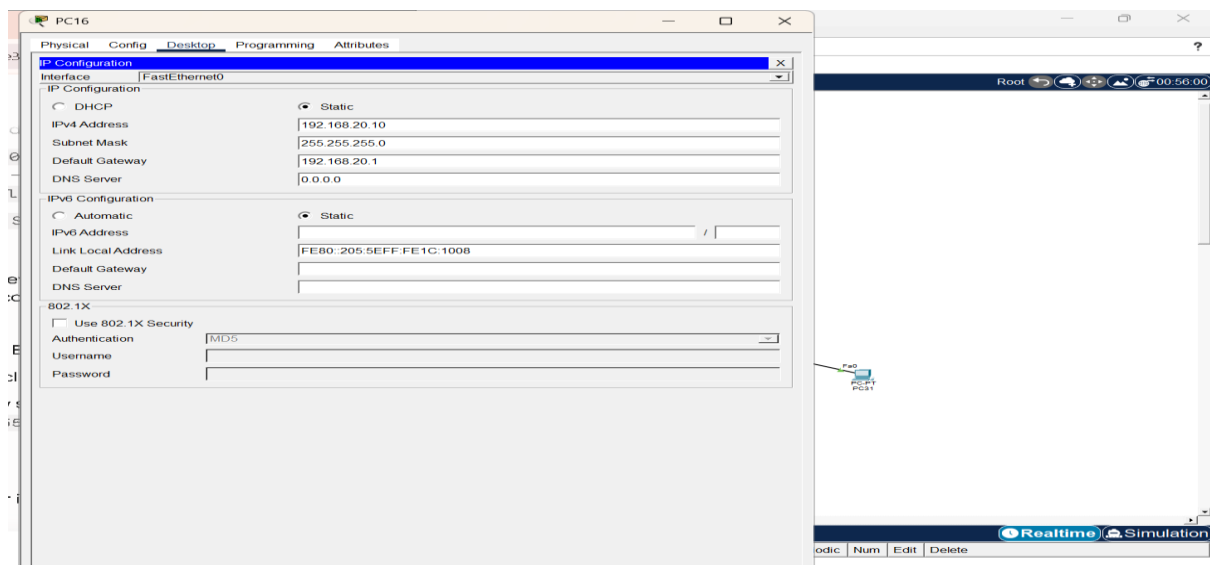
| Network   | Network Address | Prefix | Subnet Mask   | Gateway      |
|-----------|-----------------|--------|---------------|--------------|
| Network 1 | 192.168.10.0    | /24    | 255.255.255.0 | 192.168.10.1 |
| Network 2 | 192.168.20.0    | /24    | 255.255.255.0 | 192.168.20.1 |
| Network 3 | 192.168.30.0    | /24    | 255.255.255.0 | 192.168.30.1 |

#### Example PC configuration

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1



**Figure 5.1:** IP Configuration of End Device

### 5.2 Addressing Design Justification

Separate networks are used to:

- Reduce broadcast-domain size.
- Organize departments.
- Simplify troubleshooting.



- Enable routing between departments.
- Provide room for future expansion.
- Avoid IP address conflicts.

### 5.3 Routing Design

Routers are responsible for forwarding packets between different IP networks.

The routing table is verified using:

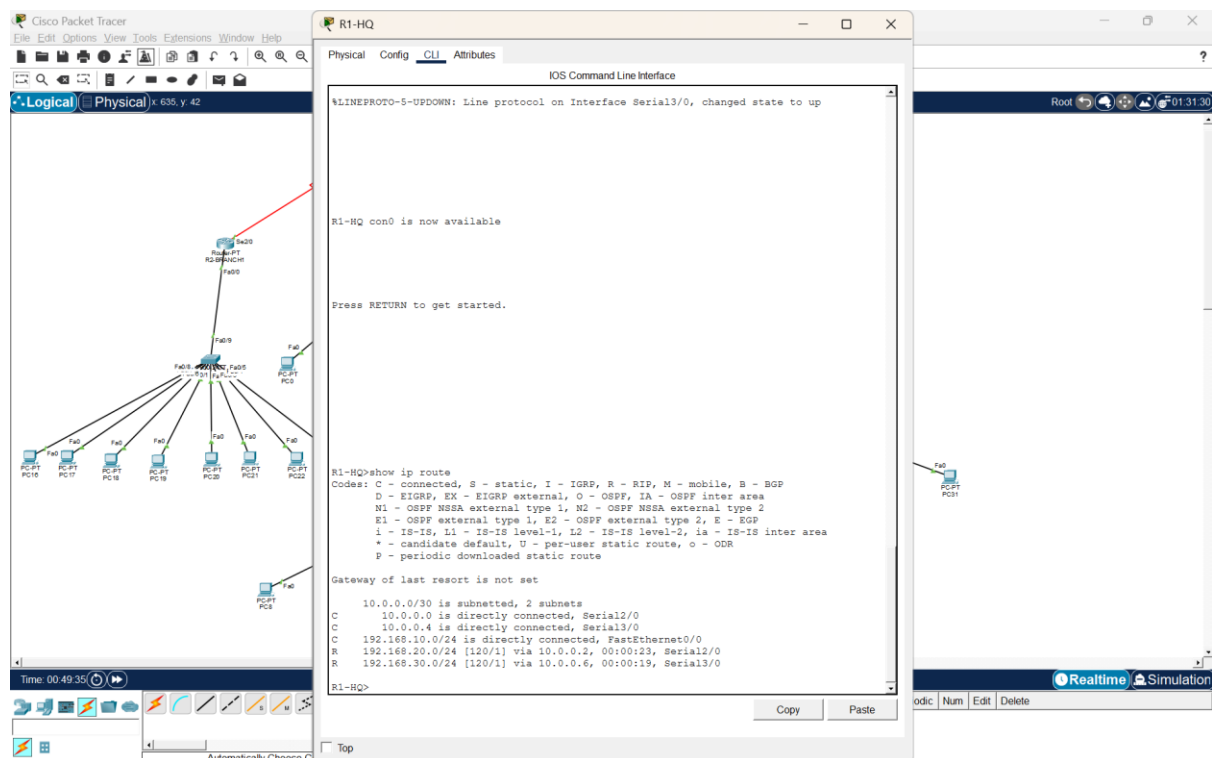
show ip route

The observed routing table contains routes for networks including:

192.168.10.0/24

192.168.20.0/24

192.168.30.0/24



**Figure 5.2: Routing Table Verification on R1-HQ**

### 5.4 Routing Verification

Routing verification checks:

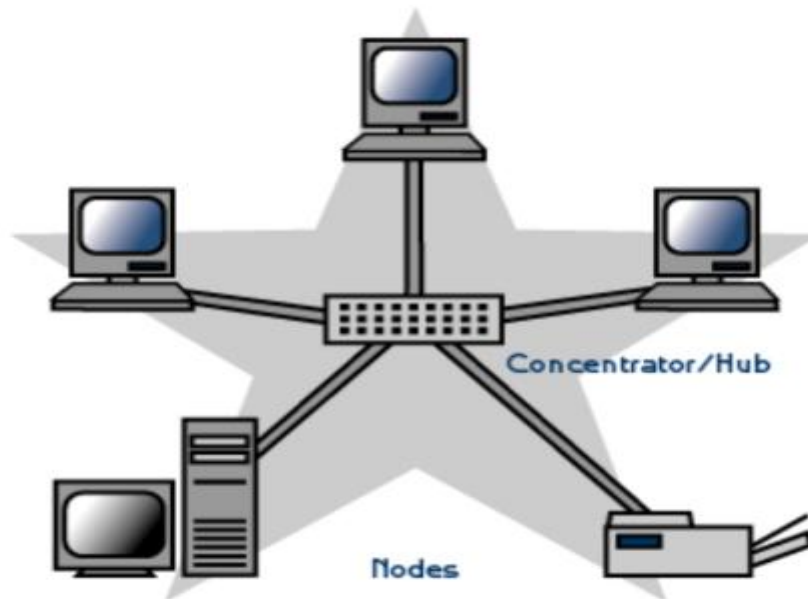
- Directly connected networks.
- Remote networks.
- Next-hop information.
- Availability of routing paths.

A successful routing table indicates that the router has learned or been configured with paths to remote networks.

## CHAPTER 6 – TOPOLOGY DESIGN AND DEVICE SELECTION

### 6.1 Star Topology

In a star topology, all end devices connect to a central switch.



**Figure 6.1:** Star Topology Structure

#### Advantages

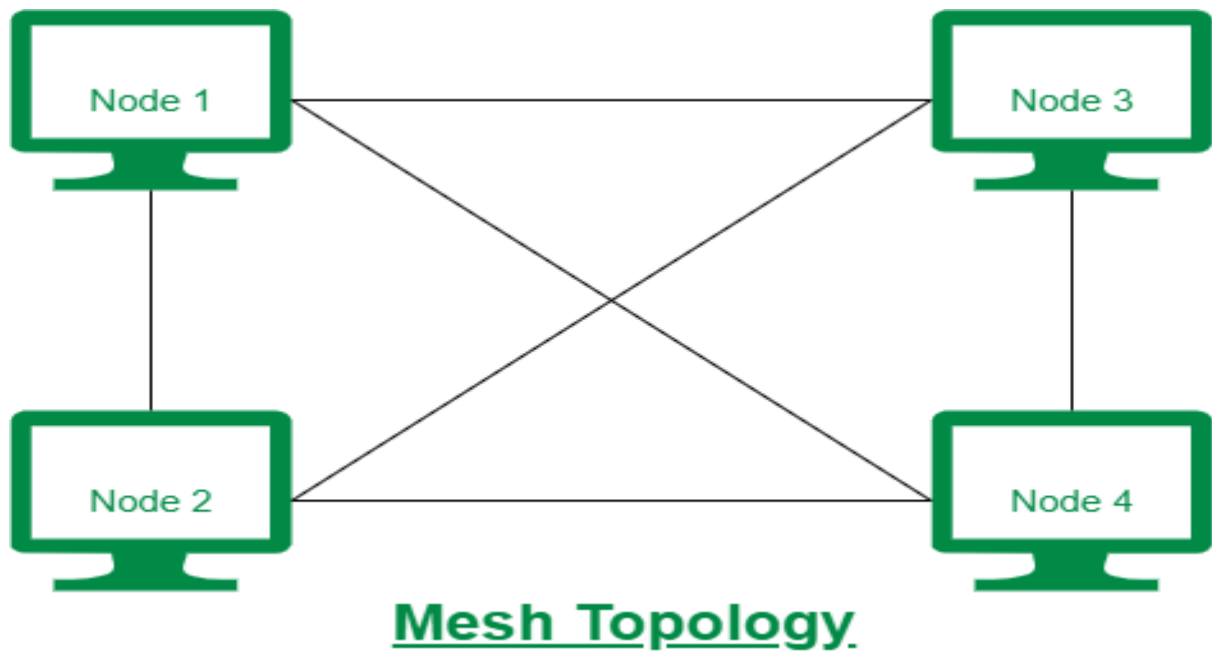
- Easy to manage.
- Easy troubleshooting.
- Failure of one PC cable affects only that PC.
- Easy to expand.

#### Disadvantage

- Failure of the central switch can affect connected devices.

### 6.2 Mesh Topology

A mesh topology provides multiple connections between network devices.



**Figure 6.2:** Mesh Topology Structure

#### **Advantages**

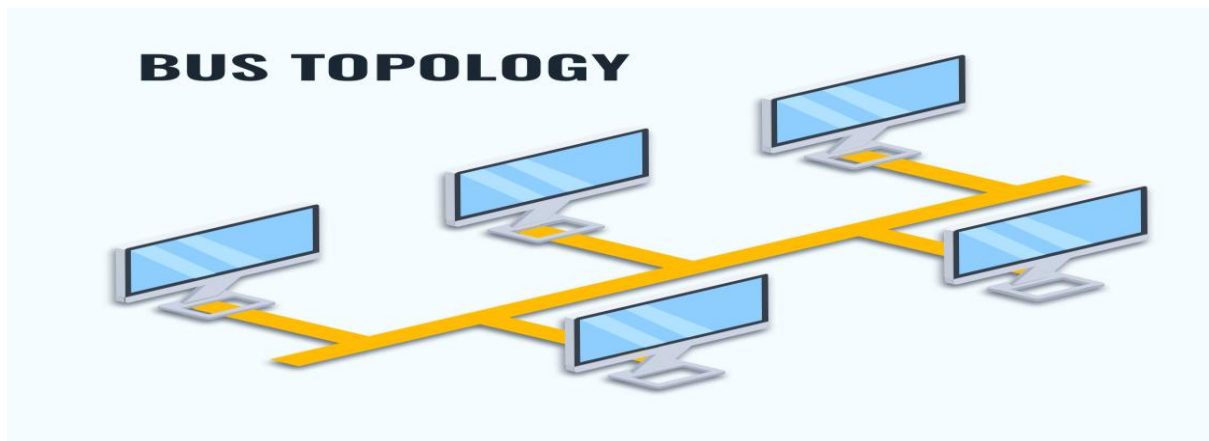
- High redundancy.
- High fault tolerance.
- Alternative paths may be available.

#### **Disadvantages**

- Higher cost.
- More cabling.
- More complex configuration.

### **6.3 Bus Topology**

A bus topology uses a shared communication medium.



**Figure 6.3:** Bus Topology Structure

### Advantages

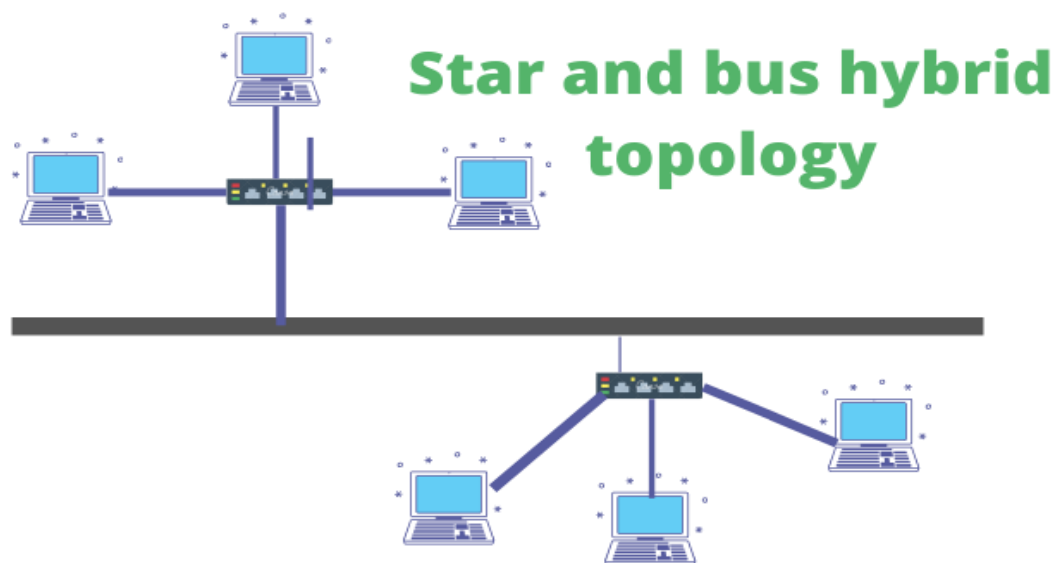
- Simple concept.
- Lower cabling requirement.

### Disadvantages

- Difficult troubleshooting.
- Shared medium can become a bottleneck.
- Poor scalability for a large enterprise.

## 6.4 Hybrid Topology

A hybrid topology combines multiple topology structures.



**Figure 6.4:** Hybrid Topology Structure

The proposed enterprise network uses a hybrid approach because it combines different network segments and connectivity structures.

### Why hybrid is suitable

- Flexible.
- Scalable.
- Supports different departments.
- Easier to expand.
- Can combine reliability and centralized management.

## **CHAPTER 7 – IMPLEMENTATION**

### **7.1 Cisco Packet Tracer Implementation**

The network was implemented using Cisco Packet Tracer.

The implementation procedure was:

1. Place three routers.
2. Place six switches.
3. Place end devices.
4. Connect routers and switches.
5. Connect PCs to appropriate switches.
6. Assign IP addresses.
7. Configure default gateways.
8. Configure routing.
9. Verify router interfaces.
10. Verify routing tables.
11. Test connectivity.
12. Observe packets using Simulation Mode.

### **7.2 Device Naming**

The devices were named according to their locations/functions:

- R1-HQ
- R2-BRANCH1
- R3-BRANCH2
- SW-HR
- SW-RD
- SW-SALES
- SW-FIN
- SW-BR1
- SW-BR2

This naming scheme improves readability and simplifies troubleshooting.

## **CHAPTER 8 – CONNECTIVITY TESTING**

### **8.1 Ping Test**

Connectivity was tested using:

ping 192.168.10.10

The ping process sends ICMP echo requests to the destination and measures whether responses are received.

### **8.2 Successful Connectivity**

One of the observed tests produced:

Packets: Sent = 4, Received = 4, Lost = 0

Loss = 0%

The observed RTT was:

Minimum = 1 ms

Maximum = 13 ms

Average = 5 ms

#### **Interpretation**

The result indicates successful packet delivery during that test. The average round-trip time of 5 ms indicates a relatively low simulated delay for the tested communication path.

### **8.3 Packet Loss Test**

Another observed test produced:

Packets: Sent = 4

Received = 3

Lost = 1

Loss = 25%

Observed RTT:

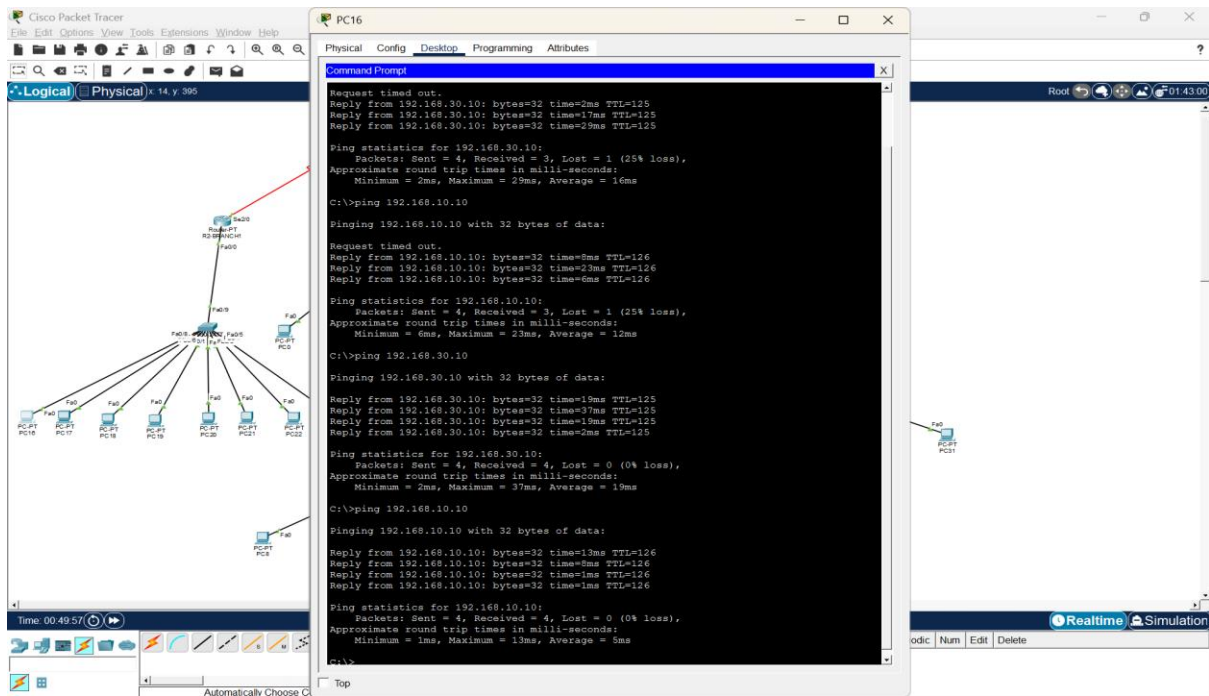
Minimum = 6 ms

Maximum = 23 ms

Average = 12 ms

#### **Interpretation**

The 25% packet loss indicates that one of the four transmitted packets was not successfully received during that test. Because this is a simulated environment, the result can be used to demonstrate how packet delivery and latency can vary during network operation.



**Figure 8.1: Ping Test and Packet Loss Results**

## CHAPTER 9 – PROTOCOL ANALYSIS

### 9.1 TCP

TCP is a connection-oriented transport protocol.

Important characteristics:

- Connection establishment.
- Reliable delivery.
- Sequencing.
- Acknowledgements.
- Retransmission.

TCP is suitable for applications where reliable delivery is important.

### 9.2 UDP

UDP is connectionless and has lower protocol overhead.

Characteristics:

- No connection establishment.
- No guaranteed delivery.
- Lower overhead.
- Faster transmission in suitable scenarios.

UDP is useful where low overhead is more important than guaranteed delivery.

### 9.3 HTTP

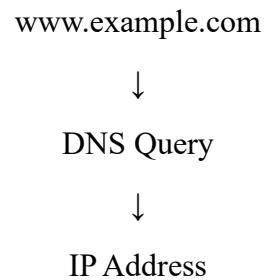
HTTP is an application-layer protocol used for web communication.

In the simulation, HTTP traffic can be generated between a client and web server and observed using Packet Tracer Simulation Mode.

### 9.4 DNS

DNS translates domain names into IP addresses.

For example:



### 9.5 Protocol Comparison

| Protocol | Layer       | Main Characteristic | Reliability             |
|----------|-------------|---------------------|-------------------------|
| TCP      | Transport   | Connection-oriented | High                    |
| UDP      | Transport   | Connectionless      | No delivery guarantee   |
| HTTP     | Application | Web communication   | Uses transport protocol |
| DNS      | Application | Name resolution     | Query/response          |

#### Important for your final report

To obtain strong marks for this criterion, you should add screenshots showing these protocols under **at least three conditions**:

1. Normal traffic.
2. High traffic.
3. Link congestion/failure.

The assignment explicitly requires these varying conditions and evidence such as screenshots, logs, packet captures, or simulator outputs.



## CHAPTER 10 – PERFORMANCE EVALUATION

### 10.1 Performance Metrics

The following parameters are considered:

#### Latency

Time taken for data to travel from source to destination.

#### Packet Loss

Percentage of packets that fail to reach the destination.

#### RTT

Round-trip time between sending a packet and receiving the response.

#### Throughput

Amount of data successfully transferred during a given period.

#### Delay

Time introduced while transmitting and processing packets.

#### Jitter

Variation in packet delay.

#### Successful Delivery Rate

Percentage of packets successfully received.

### 10.2 Observed Results

| Test   | Sent | Received | Loss | Minimum RTT | Maximum RTT | Average RTT |
|--------|------|----------|------|-------------|-------------|-------------|
| Test 1 | 4    | 3        | 25%  | 2 ms        | 29 ms       | 16 ms       |
| Test 2 | 4    | 3        | 25%  | 6 ms        | 23 ms       | 12 ms       |
| Test 3 | 4    | 4        | 0%   | 1 ms        | 13 ms       | 5 ms        |

### 10.3 Performance Analysis

The results demonstrate that network performance can change between tests.

The 0% packet-loss result indicates complete delivery of the tested packets. In contrast, the 25% packet-loss results indicate that some packets were lost.

The average RTT values also differ between tests. A lower average RTT indicates a faster response for that particular test, while a higher RTT indicates greater delay.

These observations demonstrate the importance of monitoring packet loss and delay when evaluating network performance.

## **CHAPTER 11 – NETWORK CONDITIONS**

The assignment expects testing under multiple network conditions.

### **Condition 1 – Normal Traffic**

Under normal conditions:

- Network links are operational.
- Normal packet flow is expected.
- Packet loss should be minimal.
- Latency should remain relatively stable.

### **Condition 2 – High Traffic**

Additional traffic is introduced.

Expected observations:

- Increased delay.
- Possible increase in packet loss.
- Increased congestion.
- Reduced effective performance.

### **Condition 3 – Link Failure/Congestion**

A network link is temporarily disabled or congested.

Expected observations:

- Connectivity may be interrupted.
- Packets may be dropped.
- RTT may increase.
- Routing may need to use an alternative path if one exists.
- Recovery time can be measured.

## **CHAPTER 12 – SIMULATION AND VALIDATION**

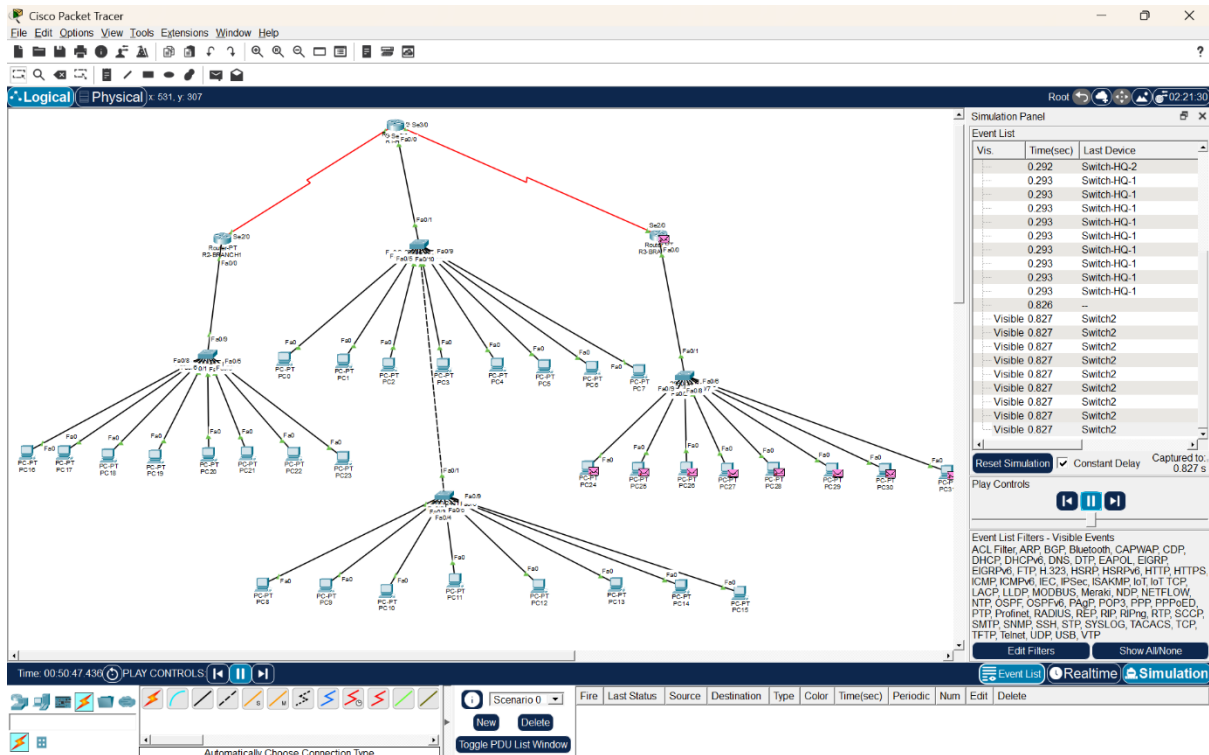
### **12.1 Simulation Mode**

Cisco Packet Tracer Simulation Mode allows packets to be observed as they move through network devices.

The Event List can be used to identify:

- Source.
- Destination.
- Network device.

- Protocol/event.
- Packet movement.
- Timing.



**Figure 12.1:** Packet Tracer Simulation Mode and Event List

## 12.2 Routing Validation

The routing table was inspected using:

show ip route

This verifies that the router has information about connected and remote networks.

## 12.3 Interface Validation

For stronger evidence, use:

show ip interface brief

This command can be used to verify:

- Interface status.
- IP addresses.
- Up/down state.
- Protocol status.

```
R1-HQ#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.10.1    YES manual up          up
FastEthernet1/0          unassigned      YES unset   administratively down down
Serial2/0                 10.0.0.1        YES manual up          up
Serial3/0                 10.0.0.5        YES manual up          up
FastEthernet4/0          unassigned      YES unset   administratively down down
FastEthernet5/0          unassigned      YES unset   administratively down down
R1-HQ#
```

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**Figure 12.2: Router Interface Verification**

## CHAPTER 13 – ENGINEERING ANALYSIS AND JUSTIFICATION

### 13.1 IP Addressing Decision

Private IPv4 addressing was selected because the simulated enterprise network does not require globally routable addresses.

Separate network ranges simplify:

- Network identification.
- Routing.
- Troubleshooting.
- Department separation.

### 13.2 Routing Decision

Routing is required because multiple IP networks are present.

Routers provide communication between:

- Head Office.
- Branch 1.
- Branch 2.
- Different network segments.

### 13.3 Switch Selection

Switches are used to connect multiple PCs within a LAN.

This is more efficient than connecting each PC directly to a router.

### 13.4 Cabling Decision

Ethernet connections are used for LAN communication, while appropriate router-to-router links are used for inter-router communication.

Automatic connection selection in Packet Tracer can simplify physical implementation, but the final report should document the actual interface and cable type used.

### 13.5 Topology Decision

A hybrid structure is appropriate for the enterprise because it provides flexibility while allowing different network segments to be organized according to their requirements.

## CHAPTER 14 – TRADE-OFF ANALYSIS

| Design Choice        | Benefit               | Trade-off                              |
|----------------------|-----------------------|----------------------------------------|
| Separate IP networks | Better organization   | Requires routing                       |
| Switch-based LAN     | Supports many PCs     | Switch failure affects connected users |
| Mesh                 | High redundancy       | Higher cost/complexity                 |
| Star                 | Easy management       | Central-device dependency              |
| Hybrid               | Flexible and scalable | More complex                           |
| TCP                  | Reliable              | Higher overhead                        |
| UDP                  | Lower overhead        | No guaranteed delivery                 |

This comparison demonstrates that network design is not based only on connectivity; cost, reliability, scalability, complexity, and performance must also be considered.

## CHAPTER 15 – LIMITATIONS AND BOTTLENECKS

### 15.1 Current Limitations

- Packet Tracer is a simulated environment.
- Simulated latency may not exactly represent a physical enterprise network.
- More extensive traffic generation is required for complete throughput analysis.
- Protocol testing needs additional experimental evidence.
- Link-failure testing should be performed for complete fault analysis.

### 15.2 Possible Bottlenecks

Potential bottlenecks include:

- Heavy traffic on inter-router links.
- Excessive traffic through a central switch.
- Single-link dependency.
- Insufficient bandwidth.
- Increased packet processing during high traffic.

## **CHAPTER 16 – SCALABILITY AND RELIABILITY**

The rubric's broader-considerations criterion specifically expects discussion of scalability, reliability, resource use, standards, safe configuration practice, and responsible network design.

### **16.1 Scalability**

The network can be expanded by:

- Adding more PCs.
- Adding switches.
- Adding departments.
- Adding branch routers.
- Allocating additional subnets.

### **16.2 Reliability**

Reliability can be improved through:

- Redundant links.
- Backup routes.
- Proper device configuration.
- Regular connectivity monitoring.
- Fault detection.

### **16.3 Resource Utilization**

Appropriate switches and subnet sizes help avoid unnecessary resource consumption.

### **16.4 Standardization**

Standard naming, addressing, and cabling practices make the network easier to maintain.

## **CHAPTER 17 – SDG RELEVANCE**

The project relates to SDG 9 because reliable digital infrastructure supports:

- Organizational communication.
- Digital services.
- Scalable IT infrastructure.
- Efficient resource utilization.
- Technology-driven business operations.

The project demonstrates how network planning and simulation can support the development of reliable and scalable digital infrastructure.

## CHAPTER 18 – RESULTS AND DISCUSSION

### 18.1 Summary of Results

The implementation demonstrated:

- Successful construction of a multi-device enterprise topology.
- IPv4 configuration of end devices.
- Default gateway configuration.
- Routing-table verification.
- End-to-end connectivity testing.
- Packet-loss measurement.
- RTT measurement.
- Packet-flow observation using Simulation Mode.

### 18.2 Key Observation

The ping experiments demonstrated that connectivity performance can vary between tests. The observed results included both **0% packet loss** and **25% packet loss**, with average RTT values ranging from **5 ms to 16 ms** in the shown tests.

These results provide initial evidence for network-performance evaluation.

## CHAPTER 19 – CONCLUSION

The project demonstrates the design and partial implementation of a multi-branch enterprise network using Cisco Packet Tracer. The network architecture includes routers, switches, and multiple end devices representing different departments and branch offices.

An IPv4 addressing scheme was applied to separate network segments, and default gateways were assigned to end devices. Routing-table verification was performed to confirm the availability of different network routes. Connectivity was evaluated using ping, and packet loss and round-trip time were recorded.

The topology analysis demonstrates that different network structures provide different levels of scalability, reliability, cost efficiency, and fault tolerance. A hybrid architecture is appropriate for the proposed enterprise environment because it provides flexibility and supports future expansion.

Further testing of TCP, UDP, HTTP, and DNS under normal, high-traffic, and failure conditions will provide more comprehensive protocol and performance analysis.

## CHAPTER 20 – FUTURE ENHANCEMENTS

Future improvements include:

1. Complete testing with all 36+ end devices.
2. Implement detailed VLSM addressing.

3. Add redundant router links.
4. Perform TCP and UDP traffic comparison.
5. Perform HTTP and DNS experiments.
6. Introduce controlled link failures.
7. Measure recovery time.
8. Measure throughput and jitter.
9. Generate performance graphs.
10. Add network security mechanisms.
11. Introduce VLAN-based department separation.
12. Implement dynamic routing for larger networks.

## CHAPTER 21 – INDIVIDUAL CONTRIBUTION

Use this format and replace the names:

| Member   | Contribution                                                        |
|----------|---------------------------------------------------------------------|
| Member 1 | Network topology design and Packet Tracer implementation            |
| Member 2 | IP addressing, subnetting, and routing configuration                |
| Member 3 | Connectivity testing, protocol analysis, results, and documentation |

The assignment specifically requires individual contribution to be clearly specified.

## CHAPTER 22 – INDIVIDUAL REFLECTION

### Reflection

During this assignment, I gained practical knowledge of enterprise network design and configuration using Cisco Packet Tracer. The most useful concept I learned was the relationship between IP addressing, subnetting, default gateways, and routing. Initially, selecting the correct interfaces and connecting routers and switches was challenging because different Packet Tracer devices provide different interface types. I learned to inspect the available interfaces before selecting the appropriate connection.

Another important learning experience was configuring end devices with suitable IP addresses and default gateways. I understood that assigning an IP address alone is not sufficient for communication with other networks; the correct subnet mask and gateway are also required.

The routing-table verification process helped me understand how routers identify directly connected and remote networks. Using commands such as `show ip route` provided practical evidence of the configured routing information.

Connectivity testing using ping was also valuable. The experiments produced different packet-loss and round-trip-time results, which helped me understand that network performance can



vary depending on network conditions. The Packet Tracer Simulation Mode further helped me visualize how packets travel between network devices.

One of the main challenges was troubleshooting connectivity when a packet did not reach the destination. Checking the IP configuration, gateway, router interfaces, and routing information provided a systematic approach to identifying the problem.

If I were to improve the design, I would introduce additional redundancy and perform more extensive protocol testing. I would also compare TCP and UDP under different traffic conditions and measure throughput, jitter, and recovery time.

Overall, the assignment improved my understanding of practical computer networking and showed me how theoretical concepts such as subnetting, routing, topology, and protocols are applied in a real network design process.

## CHAPTER 23 – REFERENCES

1. Cisco Networking Academy. *Cisco Packet Tracer*.
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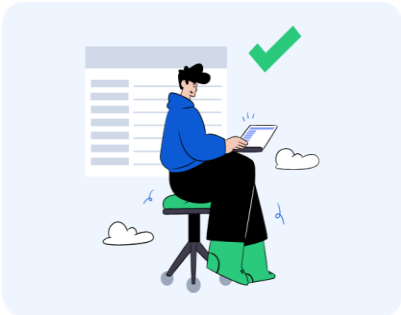
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TITLE  
Design, Configure, Analyse and Evaluate a Multi-Branch Enterprise Network Using IP Addressing, Routing, Protocols and Network Topologies for Reliable, Scalable, and Efficient Inter-Branch Communication